Vu Hoang (Gary) Pham

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Education

Drexel University, Pennoni Honors College, College of Engineering — Philadelphia, PA
M.S in Computer Engineering — Graduation: Jun 2027
B.S in Computer Science, Minors in Computer Engineering & Mathematics — Graduated: Jun 2025
Concentrations: Computer System & Architecture, Algorithms & Data Structure — GPA: 3.85 - Dean's List

Skills

Languages: C, C++, Python, Java, Racket, Haskell, Bash, Kotlin, Go, Rust, Pascal, SQL, Assembly (x86)
Tools: CUDA, CMake, VsCode, IntelliJ, Git, UNIX, CCS, Arduino, UML, DrRacket, Docker, AWS, Clang, GCC
Systems: Windows, Linux (Ubuntu, Mint, Kali, Raspberry Pi), MacOS, Plan 9, Android
Frameworks: Django, Node.js, Angular, Spring

Work Experience

Credit Genie — Plymouth Meeting, PA, US
CO-OP Backend Software Engineer — Sep 2026 - Present

- Hardened Plaid bank-connection health across 4 TypeScript services: captured error state on every API and webhook failure, and surfaced Plaid's 7-day disconnect warning to iOS/Android, so users reconnect before losing access (**1,100+** tests).
- Owned the P0 migration of an internal project-management tool into the admin-portal monorepo (**236 files, 14 PRs** currently) on Okta/Cedar auth, Postgres and EKS, and built its Snowflake employee sync.
- Built payments reliability tooling: a feature-flag kill switch blocking failing banks before they drain the Plaid rate limit, a daily cleanup job for orphaned collection attempts, and E2E harnesses against Plaid's sandbox and real iOS code.

ImesPro, Edge AI & Factory Automation — Ho Chi Minh, Vietnam
Internship Embedded Software Engineer — Jun 2026 - Sep 2026

- Developed ESP32-S3-CAM factory automation firmware in C/ESP-IDF with **ThingsBoard MQTT** (telemetry, RPC, OTA), multi-task FreeRTOS (camera, OCR uploader, health), and **double-buffered frame** upload to an OCR backend for on-line visual inspection.
- Engineered edge CV on **ESP32-S3-CAM (ESP-DL)** and **Luckfox Pico Pro (RKNN)**: trained PyTorch digit **CNN**, quantized **INT8** via **esp-ppq**, deployed on-device OCR with digit segmentation at **~10 FPS**; ran **YOLO11n** on **RV1106 NPU** with RGA letterbox at **~8–12 FPS**; shipped production QR and face firmware with dual-core **FreeRTOS**, **≥15 FPS** capture, and **MQTT** event publishing.
- Researched SmartClass inattention pipeline (YOLOv8, ByteTrack, MediaPipe) at **15-30 FPS on CPU**, exported YOLO to ONNX for Jetson TensorRT deploy, and evaluated YOLO fine-tuning on classroom datasets for tiered ESP32-gateway analytics.

Drexel University — Drexel Material Lab
Embedded Software Engineer — Nov 2025 - May 2026

- Developed an **ESP32-S3 embedded data acquisition system** in C/ESP-IDF for laser-material experiments, sampling **12-bit ADC signals at 3 kHz**, generating **5 kHz PWM** laser control, and supporting **10M-sample** capture sessions.
- Engineered a **dual-core FreeRTOS pipeline** with a **48k-entry SPIRAM queue (~16 s buffer at 3 kHz)** and **500-sample batched CSV writes**, improving logging reliability during SD-card and network I/O
- Built a **browser-based control and data delivery stack** with live telemetry, start/stop APIs, **50 s chunked downloads**, SD/SPIFSS failover, synchronized timestamps, and **30 s benchmark-mode** CSV/binary throughput testing.

Medcrypt — Solana Beach, CA, US
CO-OP Embedded Software Engineer — Apr 2024 - Sep 2024

- Designed and implemented an optimized ASN.1 encoder/decoder in C for STM32 microcontrollers, reducing the firmware footprint by **30%** (from 64 KB to 45 KB) and boosting encoding throughput by **40%**.
- Identified and resolved **150,000+ namespace symbol** conflicts in a **2 MB vendor codebase**, eliminating CI build failures and cutting integration **debugging time by 50%**.
- Unified hardware-accelerated AES encryption on the STM32 platform, slashing cryptographic **processing latency by 25%** and ensuring deterministic performance for real-time data protection.

Research

FPGA Compiler for Nested Loop Acceleration — Drexel Master Research
Hardware Researcher — Jun 25 - Present

- Designed LU decomposition and matrix factorization arrays on Xilinx FPGA (Zybo Z) using Vitis HLS and AXI-Stream IP, **achieving x2.3 speedup** over baseline C implementation.
- Compiled nested loop kernels into 64-PE arrays with **> 90% pipeline utilization** and verified through cycle-accurate simulation.
- Evaluated synthesis results via C/C++ and VHDL testbenches, optimizing DMA throughput by **35%** through bus-width and latency tuning.

Lexicographically Minimum String Rotation — Drexel Senior Project
Quantum Researcher — Sep 24 - Jun 25

- Architected and implemented a quantum LMSR algorithm in **Qiskit** using **Grover's amplification**, reducing theoretical search complexity from $O(N \log N)$ to $O(\sqrt{N})$ for N quibits this translates to **~320** fewer oracle calls.
- Simulated and deployed the **6-qubit** LMSR circuit on IBM Quantum hardware, executing **100 runs across 20 problem instances** with an average **≥ 97% success rate**, and validated against classical benchmarks.
- Applied the algorithm to benzenoid edge code recognition over **50 molecular graphs** for Drexel's Week of Excellence, slashing end-to-end **processing time by 30%**.

Activities

Participant, ICPC USA Regional Contest, Philly Code Fest, DragonHack — Varsity Player, Drexel Badminton Club, Hanoi Swimming Team